



Islamic University of Technology
Department of Computer Science and Engineering

Lab 8

CSE 4404: Algorithms Lab
Summer 2023-24

Task A. Maximum subarray sum

Time Limit: 1 second | Memory Limit: 512 MB

You are given an array of n integers. Your task is to find the maximum sum of a subarray. A subarray is a contiguous part of the array.

Input Format

The first input line has an integer n : the size of the array. The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output Format

Print one integer: the maximum subarray sum.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x_i \leq 10^9$

Examples

Sample Input	Sample Output
8 -1 3 -2 5 3 -5 2 2	9

NB: You must use a divide and conquer approach

Task B. Minimum Euclidean Distance

Time Limit: 1 second | Memory Limit: 512 MB

You are given n points on a 2D plane. What is the minimum distance between two points?

Input Format

The first input line contains an integer n : the number of points. After this, there are n lines that describe the points. Each line has two integers x and y : the coordinates of a point. All points are distinct.

Output Format

Print one integer: the minimum distance squared between two points.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Examples

Sample Input	Sample Output
4 0 0 5 5 3 4 1 1	2

NB: You must use a divide and conquer approach